THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON

1. The area function of the excursion skeleton

Convention on RH. Every statement carries one of three labels: *[unconditional]*, *[under RH]* (the Riemann hypothesis is assumed; simple zeros are stated explicitly when also needed), or *[heuristic]* (not proved; a model). The definitions and the exact identities of §1.1–§1.2 are unconditional: they use no hypothesis on the zeros of ζ.

1.1. **Definitions.** Let μ be the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad x \ge 0,$$

the Mertens function, piecewise constant with jumps at the integers. Let $0 = z_0 < z_1 < z_2 < \dots$ be the points where M vanishes. Since the jumps of M are 0 or ± 1, every sign change passes through a zero, and M changes sign infinitely often *[unconditional]*. The ν-th *excursion* is the interval

$$I_\nu = (u_\nu, v_\nu], \qquad u_\nu = z_{\nu-1}, \quad v_\nu = z_\nu, \qquad \ell_\nu = v_\nu - u_\nu,$$

in whose interior M does not vanish. Inside I_ν we set

$$H_\nu = \max_{u_\nu < n \le v_\nu} |M(n)|, \qquad \tau_\nu = \min\{n \in I_\nu : \ |M(n)| = H_\nu\}, \qquad \sigma_\nu = \operatorname{sgn} M(\tau_\nu) \in \{\pm 1\},$$

and we write $a_\nu = \tau_\nu - u_\nu$, $b_\nu = v_\nu - \tau_\nu$, so that $\ell_\nu = a_\nu + b_\nu$. Since M needs at least H_ν steps to climb from 0 to H_ν and as many to return, one has $H_\nu \le \min(a_\nu, b_\nu)$ *[unconditional]*.

Definition 1.1 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on each excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu}\, \dfrac{t - u_\nu}{a_\nu}, & u_\nu \le t \le \tau_\nu, \\[2mm] \sigma_\nu \sqrt{H_\nu}\, \dfrac{v_\nu - t}{b_\nu}, & \tau_\nu \le t \le v_\nu. \end{cases}$$

On the terminal block $(u, N]$, if $M(N) \neq 0$, the descent ends at $\sigma\sqrt{|M(N)|}$ instead of at 0. We also write L for the *unregularized* skeleton, defined in the same way but with peak $\sigma_\nu H_\nu$ instead of $\sigma_\nu \sqrt{H_\nu}$.

Definition 1.2 (The three curves)**.**

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\, dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\, dt, \qquad \mathcal{A}(x) = \mathcal{A}_+(x) + \mathcal{A}_-(x) = \int_0^x M_{\mathrm{s}}(t)\, dt,$$

where $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$. In the figure, $\mathcal{A}_+$ is the blue curve, $\mathcal{A}_-$ the red one and $\mathcal{A}$ the black one. $\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on the excursions of the opposite sign: this is what produces the plateaus.

1.2. **Exact formulas.**

Lemma 1.3 (Increment per excursion)**.** [unconditional] *For every complete excursion,*

$$\mathcal{A}(v_\nu) - \mathcal{A}(u_\nu) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}.$$

The discrete version agrees: $\sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu\sqrt{H_\nu}\big((a_\nu+1)+(b_\nu-1)\big)/2 = \sigma_\nu\sqrt{H_\nu}\,\ell_\nu/2$.

Proof. The two pieces of M_{s} are triangles of height $\sqrt{H_\nu}$ with bases a_ν and b_ν; their areas are $\sqrt{H_\nu}a_\nu/2$ and $\sqrt{H_\nu}b_\nu/2$. In the discrete sum, the ascent contributes $\sqrt{H_\nu}\sum_{j=1}^{a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu+1)/2$ and the descent $\sqrt{H_\nu}\sum_{j=0}^{b_\nu-1} j/b_\nu = \sqrt{H_\nu}(b_\nu-1)/2$. □

Proposition 1.4 (Explicit form of the three curves)**.** [unconditional] *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(u_\nu) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x-u_\nu)^2}{2a_\nu}, & u_\nu \le x \le \tau_\nu, \\[2ex] \mathcal{A}(u_\nu) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & \tau_\nu \le x \le v_\nu, \end{cases}$$

and, at the zeros of M,

$$\mathcal{A}(v_k) = \sum_{\nu\le k} \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \mathcal{A}_+(v_k) = \sum_{\substack{\nu\le k\\ \sigma_\nu=+1}} \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \mathcal{A}_-(v_k) = -\sum_{\substack{\nu\le k\\ \sigma_\nu=-1}} \frac{\sqrt{H_\nu}\,\ell_\nu}{2}.$$

Thus the three curves are piecewise quadratic, of class C^1, and are determined by the data $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$.

Lemma 1.5 (Critical points and inflections)**.** [unconditional] *One has $\mathcal{A}' = M_{\mathrm{s}}$, so the local extrema of $\mathcal{A}$ are exactly the zeros of M, and $\mathcal{A}$ is monotone on each excursion. Since M_{s} is piecewise linear, $\mathcal{A}''$ is piecewise constant and changes sign at τ_ν: the inflection points of $\mathcal{A}$ are the first argmaxima of $|M|$.*

1.3. **Where ζ enters.** The black curve depends on M nonlinearly, through the square root of the peak, so $\mathcal{A}$ has no Mellin transform algebraic in ζ. Its dependence on ζ is that of the data in Proposition 1.4: the nodes u_ν, v_ν are the zeros of M, the signs σ_ν are those of M on each excursion, and $H_\nu = \max_{I_\nu}|M|$. With the explicit formula of Titchmarsh [1, §14.27], valid *[under RH and simple zeros]*, the limit being taken along a suitable sequence $T_k \to \infty$ and x not an integer,

$$M(x) = \sum_\rho \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\ge1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{n\,(2n)!\,\zeta(2n+1)}, \tag{1}$$

the three ingredients above are written in terms of the zeros of ζ.

For the unregularized skeleton L, by contrast, the dependence is linear. If $\widetilde{A}(x) = \int_0^x M(t)\,dt$, then $\int_0^\infty \widetilde{A}(x)x^{-s-2}\,dx = \dfrac{1}{s(s+1)\zeta(s)}$ for $\Re s > 1$, and by Mellin inversion, computing the residues at ρ, at $s=0$ (with $\zeta(0) = -\frac12$) and at $s=-1$ (with $\zeta(-1) = -\frac{1}{12}$),

$$\widetilde{A}(x) = \sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + O(1/x) \qquad \textit{[under RH and simple zeros]}, \tag{2}$$

a series that converges better than (1) thanks to the factor $|\rho|^{-2}$. Moreover, with $R = M - L$ one has the unconditional identity

$$\int_0^{v_k} L = \widetilde{A}(v_k) + \sum_{\text{legs } \subseteq(0,v_k]} \sum_{m\in\text{leg}} \mu(m)\,(m - m_{\text{leg}}), \tag{3}$$

where each leg is an ascent or a descent and m_{leg} is its midpoint: the area of the residual on each leg is exactly a dipole of μ.

1.4. **Bounds.**

Lemma 1.6 (Pointwise transfer)**.** [unconditional] *If $|M(t)| \le c\,t^{1/2}$ for all $t \le x$, then $|M_{\mathrm{s}}(t)| \le c^{1/2}t^{1/4}$ for all $t \le x$, and therefore*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \le \tfrac{4}{5}\,c^{1/2}\,x^{5/4}, \qquad |\mathcal{A}(x)| \le \tfrac{4}{5}\,c^{1/2}\,x^{5/4}.$$

Proof. Let $t \in I_\nu$. If $t \ge \tau_\nu$, then $|M_{\mathrm{s}}(t)| \le \sqrt{H_\nu} \le c^{1/2}\tau_\nu^{1/4} \le c^{1/2}t^{1/4}$. If $t \le \tau_\nu$, then $|M_{\mathrm{s}}(t)|$ is the line joining $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$, and $\sqrt{H_\nu} \le c^{1/2}\tau_\nu^{1/4}$; since $t \mapsto c^{1/2}t^{1/4}$ is concave and nonnegative at u_ν, it lies above that chord. Integrating, $\int_0^x c^{1/2}t^{1/4}dt = \frac{4}{5}c^{1/2}x^{5/4}$. □

Corollary 1.7 (Unconditional bound)**.** [unconditional] *By the Vinogradov–Korobov zero-free region, $M(x) \ll x\exp\bigl(-c(\log x)^{3/5}(\log\log x)^{-1/5}\bigr)$, whence*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2}\exp\Bigl(-\tfrac{c}{2}(\log x)^{3/5}(\log\log x)^{-1/5}\Bigr).$$

Corollary 1.8 (Bound under RH)**.** [under RH] *$M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ implies $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}$. Conversely, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\varepsilon}$ along a sequence $x \to \infty$ for some fixed $\varepsilon > 0$, then RH is false.*

Remark 1.9. The converse goes only in that direction. Since $\ell_\nu \ge 2H_\nu$, a bound $\sqrt{H_\nu}\ell_\nu \ll x^{5/4+\varepsilon}$ yields only $H_\nu \ll x^{5/6+\varepsilon}$, far from RH. The black curve is a refutation test, not a criterion.

Remark 1.10 (Oscillation)**.** *[unconditional]* $M(x) = \Omega_\pm(x^{1/2-\varepsilon})$ for every $\varepsilon > 0$, because ζ has zeros with $\Re\rho = 1/2$. Moreover the Mertens conjecture is false: $\limsup M(x)x^{-1/2} > 1$ and $\liminf M(x)x^{-1/2} < -1$ (Odlyzko–te Riele [2], with later improvements). Hence $H_\nu \gg v_\nu^{1/2-\varepsilon}$ for infinitely many excursions, and neither $\mathcal{A}_+$ nor $\mathcal{A}_-$ stabilizes.

1.5. **Criteria equivalent to RH.** The two equivalences below concern the area of M and the energy of the black curve; in both, the only analytic ingredient is the L^1 Mellin criterion: if $\int_1^\infty |M(x)|x^{-\sigma-1}dx < \infty$ for every $\sigma > 1/2$, then $1/(s\zeta(s)) = \int_1^\infty M(x)x^{-s-1}dx$ is analytic in $\Re s > 1/2$, so ζ has no zeros there.

Proposition 1.11 (Excursionwise L^1 criterion)**.** [unconditional: an equivalence, RH is not assumed] *Let $\beta_\nu = \int_{I_\nu}|M|$. Then*

$$\mathrm{RH} \iff \sum_{v_\nu \le x}\beta_\nu \ \ll_\varepsilon\ x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0,$$

the excursion in progress included. The sum does not depend on the signs σ_ν: RH is a statement about the profile $|M|$.

Theorem 1.12 (Dirichlet energy of the black curve)**.** [unconditional: an equivalence, RH is not assumed] *Let $\mathcal{J}$ be the linear interpolation of $\mathcal{A}$ between consecutive critical points and*

$$E(x) = \int_0^x (\mathcal{J}')^2 = \sum_{v_\nu \le x}\frac{\bigl(\mathcal{A}(v_\nu) - \mathcal{A}(u_\nu)\bigr)^2}{\ell_\nu} + \frac{(x-u)\max_{u<n\le x}|M(n)|}{4},$$

where $(u, x]$ is the leg in progress. By Lemma 1.3, $\bigl(\mathcal{A}(v_\nu) - \mathcal{A}(u_\nu)\bigr)^2/\ell_\nu = H_\nu\ell_\nu/4$, so that $E(x) = \frac{1}{4}\bigl(\sum_{v_\nu\le x}H_\nu\ell_\nu + (x-u)H_{\mathrm{run}}\bigr)$. Then

$$\sum_{n\le x}|M(n)| \ \le\ 4E(x) \ \le\ x\max_{n\le x}|M(n)|,$$

and

$$\mathrm{RH} \iff E(x) \ll_\varepsilon x^{3/2+\varepsilon} \quad \textit{for every } \varepsilon > 0.$$

Proof. The lower bound is $|M(n)| \le H_\nu$ summed over each excursion, and the upper one is $H_\nu \le \max_{n\le x}|M(n)|$ together with $\sum \ell_\nu \le x$. If RH holds, $\max_{n\le x}|M(n)| \ll_\varepsilon x^{1/2+\varepsilon}$ and the upper bound gives $E(x) \ll x^{3/2+\varepsilon}$. Conversely, if $E(x) \ll x^{3/2+\varepsilon}$, the lower bound gives $\sum_{n\le x}|M(n)| \ll x^{3/2+\varepsilon}$; by partial summation, $\int_1^\infty |M(x)|x^{-\sigma-1}dx < \infty$ for $\sigma > 1/2$, and the Mellin criterion concludes. □

Remark 1.13. E is an average: $4E(x) = \sum H_\nu \ell_\nu$ is the excursion height weighted by ℓ_ν. It allows isolated excursions with $H_\nu \gg v_\nu^{1/2}$ provided they are short or rare, and only forbids the weighted mass from exceeding $x^{3/2+\varepsilon}$. It is moreover a sum of nonnegative local terms, with no sign cancellation.

1.6. **Model of the black curve.**

Heuristic 1.14 (Shape of $\mathcal{A}$). *[heuristic; assumes RH, simple zeros, convergence of the Fourier–Bohr expansion and a constant fill factor]* Writing $M(t) = t^{1/2}F(\log t)$ with $F(u) = 2\Re\sum_{\gamma>0} e^{i\gamma u}/(\rho\zeta'(\rho))$, and approximating $M_{\mathrm{s}} \approx \kappa\,\mathrm{sgn}(M)|M|^{1/2}$ with κ constant, one gets $M_{\mathrm{s}}(t) \approx \kappa\, t^{1/4}G(\log t)$ with $G = \mathrm{sgn}(F)|F|^{1/2}$. If $G(u) = \sum_\lambda g_\lambda e^{i\lambda u}$, then

$$\mathcal{A}(x) \approx \kappa\, x^{5/4} \sum_\lambda \frac{g_\lambda\, x^{i\lambda}}{5/4 + i\lambda}.$$

This reproduces the exponent $5/4$, the smoothness of the waves (the factor $|5/4+i\lambda|^{-1}$ damps the high frequencies) and the quasi-periodicity in $\log x$: with dominant frequency $\gamma_1 = 14.1347$, the period is $2\pi/\gamma_1$ in $\log x$, a factor ≈ 1.56 in x.

1.7. **Numerical data.**

With $N = 455\,052\,511$ and $Z(N) = 87\,385$ excursions:

$$\mathcal{A}_+(N) = 6.3449\cdot 10^9, \qquad \mathcal{A}_-(N) = -7.0877\cdot 10^9, \qquad \mathcal{A}(N) = -7.4279\cdot 10^8,$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\beta_\nu}{\ell_\nu^{3/2}} = 1.0214.$$

The bound of Lemma 1.6 with $c = \max_{10^3\le n\le N}|M(n)|n^{-1/2}$ lies about 30 times above $|\mathcal{A}(N)|$. In this range $|M(x)| \lesssim 0.5\sqrt{x}$, well inside any bound at the RH scale, so the data do not separate $x^{1/2}$ from $x^{1/2+\varepsilon}$: the factor x^ε with $\varepsilon = 0.05$ equals ≈ 2.7 here.

References

[1] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, 2nd ed., revised by D. R. Heath-Brown, Oxford Univ. Press, 1986, §14.27.

[2] A. M. Odlyzko and H. J. J. te Riele, *Disproof of the Mertens conjecture*, J. Reine Angew. Math. **357** (1985), 138–160.

[3] N. Ng, *The distribution of the summatory function of the Möbius function*, Proc. London Math. Soc. (3) **89** (2004), 361–389.